

SUNNE, Sweden

WMO: 024230

Lat: 59.8639N

Lon: 13.1166E

Elev: 109

StdP: 100.02

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.1	(c) -16.7	(d) -22.2	(e) 0.5	(f) -19.9	(g) -18.8	(h) 0.7	(i) -16.5	(j) 9.1	(k) 4.3	(l) 7.8	(m) 4.0	(n) 0.3	(o) 260	(p) 0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	26.5	18.0	24.7	17.2	22.8	16.4	19.4	24.2	18.3	22.7	17.4	21.3	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	20.9	16.7	12.0	20.0	15.7	11.3	19.1	55.6	24.3	52.3	22.8	49.2	21.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.9	5.9	5.2	-22.9	28.6	4.1	1.7	-25.9	29.8	-28.2	30.8	-30.5	31.8	-33.5	33.1
			-23.2	21.0	3.7	1.4	-25.9	22.0	-28.0	22.8	-30.1	23.6	-32.7	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-4.3	-3.6	-0.2	4.9	9.7	14.0	16.3	14.9	10.8	5.5	1.3	-2.8
	DBStd	8.29	5.91	5.10	4.65	3.23	3.56	2.65	2.73	2.55	2.97	3.77	4.15	6.04
	HDD10.0	2177	443	381	316	157	48	2	0	1	26	144	261	397
	HDD18.3	4669	701	614	574	404	267	133	77	108	227	397	511	656
	CDD10.0	568	0	0	0	3	40	120	194	154	50	6	0	0
	CDD18.3	17	0	0	0	0	0	2	13	3	0	0	0	0
	CDH23.3	311	0	0	0	0	17	68	183	43	0	0	0	0
	CDH26.7	48	0	0	0	0	1	8	37	2	0	0	0	0
(14) Wind	WSAvg	2.0	2.0	2.0	2.0	2.1	2.1	2.1	1.9	1.9	2.0	2.1	2.2	2.0
(15) Precipitation	PrecAvg	723	46	36	32	38	68	70	85	85	66	71	65	51
	PrecMax	975	116	66	80	95	135	99	198	156	135	186	179	90
	PrecMin	532	11	15	6	8	20	33	23	28	20	16	20	12
	PrecStd	104	27	14	19	23	28	19	40	42	32	40	39	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	7.7	14.3	19.7	25.4	28.1	30.0	26.5	21.9	15.7	11.4	10.0
		MCWB	5.6	4.9	8.4	11.4	15.3	18.1	18.7	18.0	16.0	12.3	9.8	8.6
	2%	DB	5.4	5.6	10.6	17.1	22.7	25.3	27.4	24.6	19.6	13.8	10.0	7.8
		MCWB	4.0	3.9	6.2	10.0	14.2	16.7	18.8	17.8	14.9	11.4	8.8	6.7
	5%	DB	4.2	4.3	8.0	14.3	20.2	23.2	25.5	22.7	17.7	12.6	8.7	6.2
		MCWB	3.2	3.0	4.6	8.5	12.8	15.8	18.3	17.0	13.9	10.7	7.6	5.2
	10%	DB	3.0	3.1	5.9	12.1	17.2	20.9	23.2	20.8	16.1	11.4	7.4	4.8
		MCWB	2.0	2.0	3.3	7.4	11.4	14.5	17.2	16.2	13.1	9.9	6.5	3.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.8	8.7	12.6	17.0	19.2	21.3	20.1	17.1	13.4	10.3	8.7
		MCDB	7.0	6.7	13.4	18.5	22.4	24.4	26.0	23.7	19.9	14.6	10.9	9.7
	2%	WB	4.4	4.3	6.6	10.5	15.2	17.9	19.8	18.7	15.7	12.2	8.9	6.8
		MCDB	5.1	5.2	10.1	15.5	20.7	23.1	25.2	22.4	18.3	13.3	9.9	7.6
	5%	WB	3.3	3.3	5.0	9.1	13.7	16.6	18.9	17.7	14.6	11.1	7.8	5.4
		MCDB	4.1	4.2	7.4	13.6	18.7	21.8	23.9	21.1	17.0	12.1	8.6	6.0
	10%	WB	2.1	2.0	3.8	7.8	12.2	15.4	17.9	16.8	13.6	10.1	6.5	4.0
		MCDB	2.8	2.9	5.6	11.3	16.4	20.0	22.2	19.9	15.7	11.4	7.2	4.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	9.0	10.9	11.1	11.2	10.8	10.4	9.6	7.0	4.9	5.5
		MCDBR	5.5	7.1	12.1	16.1	16.8	16.2	15.1	13.6	11.9	8.0	6.1	5.8
		MCWBR	5.2	5.5	8.2	9.7	9.0	8.6	7.6	7.6	7.7	6.1	5.7	5.5
	5% WB	MCDBR	5.4	6.3	10.7	14.3	13.3	14.0	13.2	11.6	10.5	6.8	5.9	5.6
		MCWBR	5.2	5.1	7.5	8.9	7.7	8.1	7.3	7.0	7.3	5.7	5.6	5.4
(40) Clear-Sky Solar Irradiance	taub		0.261	0.303	0.328	0.366	0.365	0.355	0.375	0.367	0.336	0.325	0.283	0.238
	taud		2.211	2.298	2.364	2.348	2.413	2.462	2.422	2.441	2.517	2.457	2.292	2.129
	Ebn at Noon		531	704	806	834	864	878	847	826	798	680	500	403
	Edh at Noon		41	66	85	103	103	100	102	93	73	57	38	29
(44) All-Sky Solar Radiation	RadAvg		0.29	0.90	2.43	3.81	4.87	5.60	5.00	3.92	2.61	1.16	0.38	0.19
	RadStd		0.05	0.17	0.26	0.50	0.47	0.34	0.65	0.40	0.32	0.22	0.10	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page